

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a production multi-tenant NHS eTriage backend (Node.js, MongoDB) over sensitive clinical data for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations.
- Owned reliability for the production system: added integration tests for regression protection, monitored production logs and debugged tenant-specific incidents, hardened auth and rate limiting, and remediated **20+ VAPT findings**.
- Architected a full-stack internal platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Served as primary technical POC, wrote knowledge-transfer documentation onboarding a second developer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp in C/C++**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with **git-webhook build/deploy automation** (GitHub webhook to a droplet on push), cron jobs, and Stripe payments; live **3+ years**.

Projects

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication on Linux from a naive baseline through cache-aware tiling, **AVX/SIMD** vectorization, register-aware micro-kernels, and **OpenMP**, profiling L1/L3/TLB misses with **perf** and inspecting generated assembly.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service with client-facing APIs, provider routing/fallback across backends, and pipeline instrumentation; hardened with **150+ automated tests**.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an automated verification-tooling agent that proposes and validates test cases, scores against multiple reference systems, and commits only improvements to **Git** with report cards and anti-Goodharting guardrails.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, TypeScript, JavaScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: Node.js, Express, FastAPI, React, Next.js, Git, GitHub Actions, REST APIs

Systems / Testing: Linux / command line, perf profiling, AVX/SIMD, OpenMP, pytest, integration testing, CI/CD, debugging

ML & AI: PyTorch, Hugging Face Transformers, LLM integration, RAG, NumPy, evaluation

Cloud / Platforms: AWS, GCP, Azure, Docker, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.